Local Precision and Height Contradictions in Elliptic Integer Definitions

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Abstract

We isolate strict inequalities connecting contact-height exponents, local comparison precision, and a Lucas growth argument. For the height exponent 24 in Sun's twenty-eight-unknown construction, orders $49, 13, 25$ suffice in place of $64, 16, 48$, while retaining the original degree-64 logarithm truncation. The conclusion keeps the original arithmetic hypotheses and the original number of free variables. We also record a general valuation lemma for iterated quadratic norm expressions and sharpen the final exponents of the proposed 214-coordinate quartic, conditional on its stated height supplier. No new numerical defining threshold or variable record is asserted.

1 A uniform height criterion

Here h is the logarithmic rational height. All constants in the following theorem are independent of the fixed multiplier M, the input, and the witnesses. This uniformity is an explicit hypothesis.

Theorem 1.1. *Let $q > 0$ and let k_D, k_I, k_S be positive integers satisfying*

$$k_D > 2q, \qquad k_I > q/2, \qquad k_S > q.$$

Suppose a family of rational certificates supplies $\beta \in \mathbb{Q}$, integers n, s, t, z, real $B \geq 1$, and $c = |s-1| \geq M$, with $B \geq c - 1$. Suppose its integer residual Δ and, when $\beta \neq n$, $J = |\operatorname{num}(\beta - n)| \geq 1$ satisfy

$$\begin{aligned} |\Delta| &\leq C_D B^4, \\ \Delta \neq 0 &\Longrightarrow B^2 \leq C_0(1 + |\Delta|^{q/k_D}), \\ \beta \neq n &\Longrightarrow B^2 \leq C_0(1 + J^{q/k_I}), \qquad J \leq 2Be^{h(\beta)}, \\ h(\beta) &\leq C_S c^{q/k_S}, \\ \Delta = 0 &\Longrightarrow \log B > c - 4. \end{aligned}$$

Then one sufficiently large fixed M forces $\Delta = 0$ and $\beta = n$ for every certificate in the family.

Proof. For $\Delta \neq 0$, the first two inequalities imply $B^2 \leq C_0(1 + C_D^{q/k_D} B^{4q/k_D})$. Since $4q/k_D < 2$, this bounds B by a constant independent of M. Choosing M larger than this bound plus one excludes that case.

Put $u = q/k_I < 2$. If $\beta \neq n$, then $J \geq 1$ gives $B^2 \leq 2C_0 J^u \leq 2^{u+1} C_0 B^u e^{uh(\beta)}$. Therefore

$$\log B \leq C_1 + \frac{q}{2k_I - q} h(\beta) \leq C_1 + \frac{qC_S}{2k_I - q} c^{q/k_S},$$

where C_1 is fixed. The exponent $q/k_S < 1$ makes the last expression smaller than $c - 4$ for all sufficiently large c. This contradicts the last hypothesis, since $c \geq M$. □

No coprimality between different comparison radicals is used in this criterion. Their individual divisibilities are sufficient to obtain the displayed height estimates.

2 Valuations in quadratic norm expressions

Lemma 2.1. *Let v be a discrete valuation with value group $\mathbb{Z}$, and let $v(r) = R > 0$ be odd. Evaluate a finite binary tree whose internal operation is $N_r(u, w) = u^2 + rw^2$. If its root has valuation greater than $-R/2$, assign $c = 1/2$ at the root and assign $c/2, (c+1)/2$ to the left and right children of a node assigned c. Every leaf assigned c has valuation greater than $-cR$, and $0 < c < 1$ throughout the finite tree. In particular a leaf G/r^k, with integer $k \geq 1$, satisfies $v(G) \geq k$.*

Proof. When both summands are nonzero, their valuations $2v(u)$ and $R + 2v(w)$ have opposite parity. Thus the valuation of their sum is the minimum, not a cancellation value. Zero summands have valuation $+\infty$ and cause no difficulty. A lower bound $-cR$ at a node therefore gives $v(u) > -cR/2$ and $v(w) > -(c+1)R/2$. Induction proves the assertion. Finally $v(G) > (k-c)R > k-1$, since $R \geq 1$ and $c < 1$. □

For the seven-leaf tree in [1, Section 5], the coefficients are

$$1/16,\ 9/16,\ 5/16,\ 13/16,\ 3/8,\ 7/16,\ 15/16.$$

The lemma is not restricted to that tree or to the orders $1, 48$. If the constant leaf 1 lies on the all-left path and every other leaf is integral and zero modulo the maximal ideal, the root is one modulo that ideal. This follows by induction when r is integral and supplies the corresponding nonzero condition in completeness arguments.

3 The precision change in Sun's construction

Retain all arithmetic statements consumed by [1, Sections 2–6], including the group description, the exponent-24 contact-height bound, the pole obstruction and its prepared-input converse, the simultaneous Gaussian prime-pattern theorem, and the stated real and finite-place preparation statements. These are premises of the application here, not assertions established by our finite checker.

Corollary 3.1. *In Sun's construction, keep the logarithm truncations ℓ_{64} and ℓ_{64}^{σ}, all nine point roles, the three ordinary pole predicates, and the scalar pole predicate. Change the orders of the four ratio differences and the Lucas residual from 64 to 49; change the input comparison order from 16 to 13; and change the two scalar ratio-component orders from 48 to 25. Recompute the normalizing powers with those orders, and choose the fixed multiplier using the resulting height constants. Under the same arithmetic premises, the resulting source defines exactly $\mathbb{Z}$ and has the same 28 existential unknowns, or 29 total coordinates.*

Proof. The reciprocal-abscissa tests keep order one. They place the appropriate points in the two formal groups. At least one anchor index is a unit at a consumed prime. The corresponding exact ratio and truncated ratio are integral. A positive-order ratio-difference test makes the other truncated ratio integral; the formal-logarithm unit then makes the other exact ratio integral. Only after this step is relative precision converted to absolute precision. The unchanged truncation still has precision 64, so subtracting the two anchor equations proves $L_i \equiv n_i \pmod{p^{49}}$ from the revised ratio-difference tests. The Lucas and input tests give $p^{49} \mid \Delta$ and $p^{13} \mid J$.

The ordinary norm–trace implication is independent of the chosen positive integer orders. More explicitly, at an odd contact of depth $\nu > 0$, the factors have valuations

$$\min\{0,\ 2v(D_i) + (1 - 2k_i)\nu\}.$$

They cannot cancel. The trace and pole obstruction give a place at which every such summand is greater than $-\nu/2$. Hence $v(D_i) > (k_i - 3/4)\nu \geq k_i - 3/4$, and $v(D_i) \geq k_i$. This proves the

revised individual radical divisibilities for all three scales, without any coprimality claim between them.

Lemma 2.1 gives order 25 for the scalar ratio components. The formal logarithm preserves the valuation of the scalar ratio, so its index identity yields $M_{\mathrm{sc}}^{25} \mid (s-1)$. The unchanged height supplier gives

$$B^2 \ll 1 + |\Delta|^{24/49}, \qquad B^2 \ll 1 + J^{24/13}, \qquad h(\beta) \ll c^{24/25}.$$

The implied constants remain fixed before M is chosen. Theorem 1.1 now applies, since $49 > 48$, $13 > 12$, and $25 > 24$. In particular the input estimate is $\log B \le C + 12h(\beta)$. The original Lucas implication and $\beta = 2X^2 + 3$ give soundness: integrality of β forces the reduced denominator of X to have square dividing 2, and hence to be one.

For completeness, choose the root indices before the contact primes, exactly as in the original construction. The logarithm ratios at the selected places still have precision 64, which is more than every revised requirement. Thus the nonconstant terms in the ordinary norm factors have positive valuation. At a fixed preparation prime q, with $\nu = v_q(w^2)$ and $\alpha_j = v_q(A_j)$, the lower bound remains $3\nu + (10k_i - 8)\alpha_j$, which tends to infinity with ν for each fixed scale and order. The same square-denominator and trace-unit arguments apply.

For the scalar tree, replace the local requirement $v_q(s-1) - v_q(h) > 48v_q(r_0) + 1$ by the analogous requirement with 25. The coefficientwise preparation at primes above 2 is retained. The real bounds are also uniform: if $R = |r_0| \ge 1$ and the scalar coefficients are bounded by A, the bottom pair involving the ratio now has absolute value at most $2A^2/R^{49}$. Its parent is bounded by $(A^2 + 4A^4)/R^2$, just as before. Thus the same form of input-independent real bound supplies the prepared pole inputs. All subsequent choices are made in the original order.

Only integer exponents in rational formulas have changed. The finite point and pole-root choices and the twenty-seven free rational parameters are unchanged. The same all-specializations elimination argument adds one existential coordinate. No generic-open restriction or extra free coordinate has been introduced. □

The three orders are the least positive integers satisfying the three strict inequalities of Theorem 1.1 for $q = 24$. They are not lower bounds for every conceivable argument. We retain ℓ_{64} deliberately: reducing its degree is not needed here. No numerical total-degree bound or expanded coefficient table for the modified polynomial is claimed.

4 Keeping the stronger exponent in the 214-coordinate argument

Consider the construction at upstream revision `c84bbfa4dbe8f32fd3bf29df029f7b7849d36202`, specifically `research/quartic214/proofs/uniform-existence.md`, P08–P12. This section is a consequence of its explicitly stated input estimates, not a new verification of their geometric or analytic foundations.

Lemma 4.1. *Suppose two contact blocks have radicals $M_1, M_2 \ge 1$, disjoint outside a fixed finite set, and both give $h(t) - C_0 \le C_1 M_i^{\alpha}$ when $h(t) > C_0$. Then $h(t) \le C_0 + C_1(M_1M_2)^{\alpha/2}$. Consequently, if two comparison scales separately have radical M_j with $M_j^k \mid J \ne 0$, and the aggregate trace comparison is $B^2 \le Ah(T) + A'$, then*

$$B^2 \le C|J|^{\alpha/(2k)} + A'$$

for a fixed C, after adjusting the fixed additive constants.

Proof. Multiply the two positive inequalities and take square roots. For the scales $t_0 = \lambda$, $t_1 = T\lambda$, use $h(T) \le h(t_0) + h(t_1)$ and the two separate divisibilities. The case $h(t) \le C_0$ is included by the additive constant. □

The quoted supplier gives $\alpha = 2103/1024$; replacing it by $7/3$ is unnecessary. Lemma 4.1 gives

$$h(t) \ll M^{2103/2048}, \qquad \Delta \neq 0 \Longrightarrow B^2 \leq C' B^{701/512} + C'',$$

because $k = 3$ and $|\Delta| \leq 14B^4$. The power needed when solving the last inequality is

$$\frac{1}{2 - 701/512} = \frac{512}{323},$$

in place of $9/4$. For the input comparison, $k = 1$ and $J \leq 2Be^{h(\beta)}$ give, above the fixed additive-height threshold,

$$\log B \leq C + \frac{2103}{1993} h(\beta),$$

in place of $C + (7/5)h(\beta)$. These improvements change no coordinate or quadratic row. They do not remove the ineffective aggregate-height constant. Since $\alpha > 2$, order two still does not satisfy the strict Lucas inequality in this argument.

5 A multiplicity-preserving height average

The product construction in the proof of [2, Lemma 16.2(1)] keeps repeated simple factors. The following elementary consequence explains the precise use needed by the quartic height argument.

Lemma 5.1. *Let J be a g-dimensional abelian variety over a number field L, and let A be simple over L, of dimension a. Suppose J contains an L-subvariety L-isogenous to A^r, $r \geq 1$. Assume there is an L-isogenous variety J_0, isomorphic over L to a product of simple varieties, with $h_F(J_0) \leq h_F(J)$. Assume also $h_F(B) \geq -c \dim B$ for all remaining product factors, with the same height normalization. Then some L-simple factor C, L-isogenous to A, satisfies*

$$h_F(C) \leq \frac{h_F(J) + cg}{r} - ca.$$

Proof. Semisimplicity up to L-isogeny gives at least r occurrences of the simple isogeny class of A in J_0. Select r occurrences, even if some factors are isomorphic, and write $J_0 = C_1 \times \cdots \times C_r \times E$. Then $\dim E = g - ra$. Height additivity and the lower bound give $\sum_i h_F(C_i) \leq h_F(J) + c(g - ra)$. At least one factor is no larger than the average. No invariance under arbitrary isogeny is used. □

For surfaces and $c = 3/2$ this retains a subtractive term 3 compared with the coarser bound $(h_F(J) + 3g/2)/r$. One must still transfer height from the selected factor to the original A by an isogeny in the correct direction and pay its half-logarithmic degree cost.

6 Verification and scope

The exact checker verifies the rational exponents, the norm-tree coefficients, and the valuation residue classes of the revised quartic factors. It rejects the three boundary precision choices $48, 12, 24$ for the strict criterion. Its small division-polynomial and scalar congruence tests are explicitly finite evidence, not universal proofs. The source constants in the two applications are different; they are not identified. The product-height result in the companion paper and the sum-height input used by the quartic are likewise distinct.

References

[1] Zhi-Wei Sun, $\mathbb{Z}$ *is Diophantine over* $\mathbb{Q}$ *with 28 unknowns*, arXiv:2611.14605v1, 10 November 2026, https://arxiv.org/html/2611.14605v1.

[2] Éric Gaudron and Gaël Rémond, *Nouveaux théorèmes d'isogénie*, Mémoires de la Société Mathématique de France 176 (2023), DOI: 10.24033/msmf.484. The directly inspected text for the product construction is the authors' September 2022 version, Lemma 16.2(1), printed p. 104, https://lmbp.uca.fr/~gaudron/art20.pdf. The final published page comparison is not certified here.